Assignment 1: Reactive Internet Application

Introduction

The objective of this **individual assignment** is to develop an Internet Application in CSU44000, which demonstrates the following features.

Client implementation requirements

- A reactive client running in an Internet Browser using Vue.js.
- The client must utilize the V- directives and moustache syntax.
- The client must interact with a Server-Side application using at least one Web API primitive.
- You may not use Vue-CLI or any other 'scaffolding' system.

The client should satisfy all client-side requirements. However, the students may choose to implement the client in React. In this case, the client must demonstrate reactive functionality through the use of components, props (properties), state (useState), and event handling, and provide the functionality equivalent to the Vue directives (e.g., conditional rendering in React vs. v-if in Vue, and list rendering in React vs. v-for in Vue).

Server implementation requirements

A Server-side application which exposes at least one API primitive and consumes the services of a 3rd party Web API (you won't be able to do this from the client side due to 'same-origin' restrictions).

Application details

You are asked to produce an application that inputs the city that someone is planning to go to. Your server-side should use the Openweathermap API to get the weather forecast for that city for the next 3 days. The client side should then display some summary information, including:

- Packing: if there is rain anytime over the next 3-days indicate that the user should bring an umbrella.
- Indicate the type of whether that the user should pack for Cold (temperatures below 8C), Mild (temperatures from 8C to 24C inclusive) or Hot (temperatures above 24C).
- Give a summary table for the next 3 days showing: Temperature, Wind Speed and Rainfall level.

The Openweathermap API is documented here: https://openweathermap.org/api.

When you request the weather data for a city, it will return you the Longitude and Latitude of the city. You can then use these in the AIR Pollution API to get detailed information on a number of different air pollutants. If any of these pollutants exceed the categorised as "Good" in the API documentation a warning message should be generated detailing which pollutant has been measured at elevated levels, in indication of how elevated the levels are together with the potential health risks associated with those levels.

On completion of these core features, you should implement one innovative and creative feature which requires that it is unique to your Internet Application. This innovative feature should involve a non-trivial addition or modification of both the client and the server.

As part of the assignment, you should clearly document the design process and technologies used to develop you Internet Application with a technical report of no more than 1500 words (i.e., roughly 3 pages of text). The use of figures, diagrams, or tables is highly encouraged, an each is equivalent to 100 words. Moreover, you need to demonstrate all of these features working with a video of no more than 3 minutes.

Lab Execution

You are expected to work on this assignment during your scheduled lab session “Lab 6” and outside of the lab sessions. It is worth noting that you may not collaborate as it is an individual assignment.

Submission and Grade

Please submit the following deliverables via Blackboard **before 11:59pm on Friday 23rd of October 2026**. To avoid any potential troubles with the last-minutes submissions, students are recommended to submit their assignment well in advance of the deadline.

- D1: All source code in its original format.
- D2: A single report document in pdf with all source code.
- D3: A technical report detailing the design process and technologies used to develop your Internet Application.
- D4: A video of under 3 minutes duration demonstrating all features of your Internet Application.

It is worth noting that each student can submit the assignment up to three times; however, only the last submission will be graded. It is noted that disabled students with a LENS report have a default one-week extension in accordance with TCD’s Late Submission of Assessments (Extensions) Policy. A penalty of 20% per day will apply to all other late submissions. In all cases, 0% will be awarded if the submission is more than 1 week after the original submission deadline.

You should include a declaration document stating that you commit to the TCD’s Academic Integrity Policy and **all the material submitted is your own work**. This declaration document

can be combined as part of D2 or submitted as a separate deliverable.

This assignment is worth a total of 40% of your year-end results for the CSU44000 module. The marking scheme is defined as follows.

Aspect [weight]	INCOMPLETE [0-39]	COMPETENT [40-69]	PROFICIENT [70-100]
Design Process Detailed in the Report [15%]	The student failed to demonstrate a systematic and professional approach to designing the Internet Application.	The student demonstrated a systematic and professional approach to solving the problem. This design process has been documented in the report. The design process was informed by topics introduced in the module.	The student demonstrated exceptional creativity and self-directed learning in their approach to designing the Internet Application. This design process has been thoroughly and accurately documented in the written report.
Technical Knowledge Demonstrated in the Report [20%]	The student failed to demonstrate thorough understanding of the technical topics introduced in the module in the technical report.	The student demonstrated thorough understanding of the technical topics introduced in the module in the technical report.	The student demonstrated the ability to critically evaluate tools and design techniques introduced in the module. The student has demonstrated the ability to acquire new technical skills related to the topics introduced in the module.
Client Programme [15%]	The client program contains one or more errors in the execution of the core functionality. **Or** Functionality that has been specified for the Client has been implemented on the Server.	The core functionality of the client program is executed correctly. The code has been written with good style using a good choice of variable and function names. A reasonable attempt has been made at adding innovative features.	The core functionality of the client program is executed correctly. The code has been written with excellent style using an excellent choice of variable and function names. Appropriate comments have been used. An excellent attempt has been made at adding innovative features which are unique. The functionality has been excellently presented with a video

			of under 3 minutes in length.
Server Programme [15%]	The server program contains one or more errors in the execution of the core functionality. **Or** Functionality that has been specified for the Server has been implemented on the Client.	The core functionality of the server program is executed correctly. The code has been written with good style using a good choice of variable and function names. A reasonable attempt has been made at adding innovative features.	The core functionality of the server program is executed correctly. The code has been written with excellent style using an excellent choice of variable and function names. Appropriate comments have been used. An excellent attempt has been made at adding innovative features which are unique. The functionality has been excellently presented with a video of under 3 minutes in length.
Visual Impact [15%]	The Internet Application has a poor Visual Impact when used. **Or** The video demonstrating the Internet Application is missing or of low quality or incorrect length.	The Internet Application functions with a good quality interface. This is demonstrated by a good quality video of under 3 minutes and over 2 minutes.	The Internet Application is visually appealing in intuitive to use. This has been demonstrated by a high quality video of under 3 minutes.
Report presentation and writing [5%]	Writing is of a poor standard. Has a lack of consistency. Makes insufficient or incorrect use of figures, diagrams, and tables.	The report is coherent, internally consistent, well organized and lucidly argued. Suitable use of figures, diagrams, and tables is made.	The report is excellently written, coherent, internally consistent, well organized and exceptionally argued. Excellent use of figures, tables, and diagrams where appropriate.
Innovation [15%]	The Internet Application has failed to include any	The innovative features are of superficial complexity	Innovative features of non-trivial complexity have been designed and developed. These

	functioning innovate features.	or utility or are not very innovative.	features work reliably. The functionality of these innovative features has been excellently presented with a video of under 3 minutes in length.